



Can interoceptive accuracy influence maximal performance, physiological and perceptual responses to exercise?

Daniel Gomes da Silva Machado^{a,1}, Luiz Fernando de Farias Junior^{b,1},
Paulo Henrique Duarte do Nascimento^b, Marília Padilha Martins Tavares^b,
Samara Karla Anselmo da Silva^b, Pedro Moraes Dutra Agrícola^a,
Luiz Inácio do Nascimento Neto^{a,c}, André Igor Fonteles^{d,e}, Hassan Mohamed Elsangedy^a,
Li Min Li^f, Alexandre Hideki Okano^{c,g,*}

^a Dept. of Physical Education, Faculdade Uninassau, Natal, RN, Brazil

^b Dept. of Physical Education, Federal University of Rio Grande do Norte, Natal, RN, Brazil

^c Graduate Program in Physical Education, State University of Londrina, Londrina, PR, Brazil

^d Dept. of Physical Education, University of Fortaleza, Fortaleza, CE, Brazil

^e Dept. of Physical Education, Federal University of Ceará, Fortaleza, CE, Brazil

^f Dept. of Neurology, State University of Campinas, Campinas, SP, Brazil

^g Center of Mathematics, Computing and Cognition, Universidade Federal do ABC, São Bernardo do Campo, SP, Brazil

ARTICLE INFO

Keywords:

Interoception
Ratings of perceived exertion
Heart rate variability threshold
Heartbeat detection task
Physical effort

ABSTRACT

Interoception has been shown to influence self-regulation of effort and perceived exertion during exercise. However, whether interoceptive accuracy influences submaximal and maximal exercise performance, as well as psychophysiological responses to it, remains elusive. We assessed poor ($n = 15$) and good ($n = 17$) heartbeat perceivers young men accordingly with their interoceptive accuracy. Heart rate variability (HRV) and blood pressure were measured at rest, and peak power, ratings of perceived exertion (RPE), and HR during a maximal incremental test in a bicycle ergometer. **Results:** At rest, HR, diastolic and mean blood pressure was lower, and inter-beat intervals were longer for good heartbeat perceivers, with no difference in HRV. During exercise, good and poor heartbeat perceivers exhibited the same submaximal and maximal power, HR, RPE at submaximal intensity, and physical fitness. Interoceptive accuracy does not influence cardiac autonomic modulation perceptual responses and performance at submaximal and maximal intensities during maximal incremental exercise.

1. Introduction

Interoception may be conceptualized as the sense of the physiological condition of the entire body [1], which is distinct from proprioception (i.e., sense of the body position in space) and exteroception (i.e., the perception of external environment). Craig [1,2] has extensively examined the relationship between interoception and the brain, as well as how interoception could provide a foundation for subjective feelings, emotion, and self-awareness. The neuroanatomy of interoception shows that small-diameter afferents and lamina I neurons are distinct from the large-diameter exteroceptive and proprioceptive afferents, which projects to the deep dorsal horn [1]. These

interoceptive fibers are sensitive to changes in various physiological conditions such as temperature, mechanical stress, local metabolism (acidic pH, hypoxia, hypercapnia, hypoglycemia, hypo-osmolarity and lactic acid), cell rupture (ATP and glutamate), cutaneous parasite penetration (histamine), mast cell activation (serotonin, bradykinin and eicosanoids), and immune and hormonal activity (cytokines and somatostatin) [1]. These afferent signals ascend to the brain and are re-represented in the anterior insula (a crucial site for interoceptive processing) where it is processed into a subjective evaluation of interoceptive state (i.e., re-representation). This is, in turn, forward to the orbitofrontal cortex, where hedonic valence is represented that will influence emotion, mood motivation, and consciousness (see Craig

* Corresponding author at: Universidade Federal do ABC (UFABC), Center of Mathematics, Computing and Cognition (CMCC), Bloco Delta, Sala 257, Alameda da Universidade, 3 - Anchieta, São Bernardo do Campo, São Paulo CEP 09606-070, Brazil.

E-mail addresses: limin@fcm.unicamp.br (L.M. Li), emaildookano@gmail.com, alexandre.okano@ufabc.edu.br (A.H. Okano).

¹ The first two authors contributed equally to this work.

<https://doi.org/10.1016/j.physbeh.2019.02.038>

Received 8 January 2019; Received in revised form 27 February 2019; Accepted 27 February 2019

Available online 28 February 2019

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[1,2] for a deeper review on the neuroanatomy of interoception).

Recent studies have proposed how interoception may play a role in central fatigue in endurance exercise [3,4]. Robertson and Marino [4] proposes that afferent feedback the body systems arrive at the anterior insular cortex (AIC), anterior cingulate cortex (ACC) and orbitofrontal cortex (OFC) where this information is interpreted in the context of emotional and motivational response. Finally, these afferent signals are forwarded to the lateral prefrontal cortex (LPFC) which would make a decision on stop or continue (e.g., increase, decrease or maintain pace) and then influence motor output via the premotor area and basal ganglia. More recently, McMorris et al. [3] proposed an interoceptive model of fatigue in which the AIC would receive both predictions of the afferent feedback (i.e., corollary discharge) sent by the dorsolateral prefrontal cortex (DLPFC) and compare to the actual afferent feedback from the whole body (i.e., interoceptive signals) via the lateral spinothalamic and nucleus of the solitary tract pathways. This comparison would result in a current state of awareness which would also be forwarded to the LPFC, where the decision regarding continue or stop would be made [3]. Also, this interoceptive pathways would interact with levels of catecholamine neurotransmitters, which would be crucial in motivation, and that a change from phasic to tonic firing would be a marker of central fatigue [3]. Both models agreed that interoceptive predictions or interoceptive signals are influenced by aspects such as perceived importance, competitor, goals, personality, among others, which may or may not override the interoceptive signals from the body [3,4].

Interoception has been defined and measured in various ways, which sometimes makes it difficult to compare studies. In this regard, Garfinkel et al. [5] have proposed a three dimensional model of interoception, which consists of the *interoceptive accuracy* (which is the process of accurately detecting and tracking internal bodily sensations), *interoceptive sensibility* (which is the subjective measure of an individual belief in their interoceptive ability), and *interoceptive awareness* (which is a metacognitive measure and refers to the correspondence between objective interoceptive accuracy and subjective report). Although different measures exist, the heartbeat detection task is the most predominant method for measuring interoceptive accuracy [5]. One such measure is the heartbeat tracking task, proposed by Schanry [6], which consist on the individual silently counting their heartbeats for different periods (e.g., 25, 30, 35, 40, 45, and 50 s), without manually checking. During the same period, the individual's actual heartbeat are measured. For each time interval, the individual's heartbeat count and the actual count are compared, and the accuracy score is averaged over the trials. This is percentage represent their interoceptive accuracy (i.e., % of beats they could count correctly) [5,6].

At rest, Pollatos and Schandry [7] showed that individuals with high interoceptive accuracy (i.e., good heartbeat perceivers) display higher heartbeat-evoked potential amplitude in the brain at the right central location, as well as a positive correlation between heartbeat perception accuracy and heartbeat-evoked potential, showing that different levels of interoceptive accuracy are reflected in brain activity. Besides, studies have shown a positive correlation between interoceptive accuracy and anxiety, depressive symptoms, and negative and positive affective experience [8–12]. Furthermore, Pollatos et al. [8] found that individuals with high cardiovascular reactivity (systolic and diastolic blood pressure, heart rate, stroke volume, cardiac output, total peripheral resistance, pre-ejection period) to physical stress (i.e. two minutes of a sustained 30% isometric handgrip) also exhibit a higher degree of interoceptive accuracy. They suggested that a greater sympathetic outflow would be one variable contributing to the establishment of high interoceptive accuracy [8]. Contrasting, the authors did not find the difference between individuals with high and low cardiovascular reactivity at rest for heart rate, total peripheral resistance, stroke volume, cardiac output, pre-ejection period, systolic or diastolic blood pressure [8].

Regarding dynamic exercise, which involves a greater muscle mass

and may provide a stronger physiological stimulus as compared to isometric exercise, few studies were performed that may implicate in interoception. In the early eighties, Pennebaker and Lightner [13] found out that ratings of perceived exertion (RPE) was higher when individuals ran on a treadmill with constant speed listening to the amplification of their breathing (i.e., focus on interoceptive signal) as compared to when they listened to distracting sounds (i.e., streets sounds) or no sound. On a second experiment, subjects who jogged in cross-country course (i.e., 1800-m trail through a wooded area adjacent to several sports fields; likely with more external information) finished in lower time and similar RPE compared to when they (ran the same distance on a lap course (i.e., nine circular 200-m laps around a small open field). However, interoceptive accuracy was not taken into account [13]. More recently, Herbert et al. [14] demonstrated that subjects with high interoceptive accuracy during a 15-min self-paced cycling exercise covered a shorter distance with lower heart rate, stroke volume, and cardiac output but reported similar RPE than those with low interoceptive accuracy (i.e., poor heartbeat perceivers). Taken together, these preliminary studies showed that directing the individual's attention to interoceptive signals or their level of accuracy in perceiving it may change exercise behavior.

As previously demonstrated, it has been proposed that the brain is crucial in the processing and integration of the afferent feedbacks that are consciously translated into RPE [15–19]. In fact, RPE is positively correlated with heart rate, oxygen uptake, blood lactate, and core temperature [20–24] and seems to play a key role in exercise performance [17–19,24]. However, RPE itself may be influenced by the way individuals re-represent (i.e., interpret) these interoceptive signals [1,2]. For instance, the accuracy in detecting heartbeats at rest influences how much effort individuals spend during exercise [14]. Additionally, a pioneer study in which individuals cycled inside a functional magnetic resonance imaging scan found increased activation of precentral and postcentral gyrus compared to rest [25]. Moreover, the cingulate gyrus was activated when RPE was ≥ 15 (i.e., “hard” on the Borg 6–20 points scale) as compared RPE < 15 [25]. Interestingly, the somatomotor and cingulate cortices showed increased activity during an interoceptive task (i.e., heartbeat discrimination: perceived synchrony between heartbeats with an external stimulus) [9]. This suggests that exercise seems to activate similar brain regions involved in the processing interoceptive awareness, particularly when exercise pass a given intensity (e.g., “hard,” 15 points on the 6–20 Borg RPE scale) likely due to increased afferent signaling during exercise.

At rest, the cardiovascular system is modulated predominantly by parasympathetic activity. During exercise, however, heart rate gradually increases due to the diminishing vagal modulation until its complete withdrawal. At this point, further increases in heart rate are driven by greater sympathetic outflow [25–27]. Parasympathetic withdrawal during exercise coincides with other important physiological transition points such as lactate and first ventilatory thresholds, which represents one important point in which exercise becomes moderate [25–27]. Beyond this point, there is a strong increase in interoceptive signals, not only the cardiovascular cues but also muscular, temperature, respiration, etc. Parasympathetic withdrawal during exercise can be measured noninvasively with a heart rate monitor through an index called heart rate variability threshold (HRV_{TH}) [25–27]. Thus, once individuals with high interoceptive accuracy present increased cardiovascular reactivity to physical stress [8], it would be possible that during dynamic exercise these subjects could have parasympathetic withdrawal at an earlier point, reaching HRV_{TH} prior than the low interoceptive accuracy individuals (i.e., lower workload).

During a maximal incremental exercise, the individual is submitted to a progressive load experiencing different exercise intensities until they reach volitional fatigue. This provides a greater physiological stimulus as compared to moderate intensity self-paced exercise, with higher increases heart rate, systolic volume, blood pressure, muscle activity, lactate production, ventilation, and oxygen uptake [20–24],

but even greater if compared to isometric exercise (e.g., handgrip exercise). All these changes lead to a strengthening of afferent feedback from the body's system to the brain, which in turn influences the formation of RPE [17–19,24]. So far, however, there is no information regarding the influence of interoceptive accuracy on physical performance during a whole-body maximal intensity dynamical exercise. Thus the purpose of this study was to compare individuals with high and low interoceptive accuracy regarding (a) cardiac autonomic nervous control at rest and during dynamic exercise and (b) exercise performance and ratings of perceived exertion (RPE) at submaximal and maximal intensity. This data may help to clarify whether subjects with high interoceptive accuracy have an increased sympathetic outflow [8] and a more finely tuned behavioral self-regulation of physical load [14], and whether individuals with low interoceptive accuracy tend to underestimate their physical effort [14]. Based on previous research involving exercise, behavior and interoception [8,9,12–14], it was hypothesized that subjects with high and low interoceptive accuracy could have different maximal performance, due to the difference in the perception of their body's symptoms (e.g., respiratory rate, muscle acidosis, heart rate, etc.), and different physiological and perceived exertion responses.

2. Methods

2.1. Participants

The sample size was based on a priori sample size calculation performed based on the results of a previous study [14]. We used Gpower software version 3.1.9.2 (Universität Kiel, Kiel, Germany) to calculate the adequate sample size for an independent *t*-test with the following criteria: (a) power = 0.8; (b) effect size $d = 0.95$ (which was calculated following the results of Herbert et al. [14]); (c) α error probability = .05; and (d) allocation ratio $N_2/N_1 = 1$. Accordingly, the sample size required to achieve 80% power was 15 participants in each group (30 in total). Thirty-two young males, whose characteristics are depicted in Table 1, were enrolled in this study. Participants were undergraduate students who volunteered to participate in. Inclusion criteria were to be insufficiently physically active (perform exercise in less than 5 days/week and < 30 min/day) and have no disease previously reported. Exclusion criteria were to miss any assessment session. The Institutional Review Board of Ethics Committee approved this study (N. 488/110; CAAE 13964913.8.0000.5537). After a detailed explanation of the experimental procedures, benefits, and possible risk, written informed consent was obtained from all volunteers.

2.2. Experimental design

Subjects were assessed in one visit to the laboratory. First, subjects read and signed the informed consent, and answered the Physical Activity Readiness Questionnaire (PARQ). Second, anthropometrics and cardiovascular, as well as the heartbeat perception task were measured at rest. Then, subjects performed a maximal incremental exercise test to assess the load at heart rate variability threshold and peak power. All subjects were assessed between 8 and 11 AM, in a silent room, and under similar environmental conditions (25 °C and 70% relative humidity). Participants were told to refrain from exercise and not to drink

alcohol or caffeine-containing products during the 24 h before the exercise trials.

2.3. Anthropometrics

Body mass and height were measured using an electronic scale with an attached stadiometer (H110, Welmy, São Bernardo do Campo, SP, Brazil) from which the body mass index (BMI) was calculated as the ratio between body mass and the height squared.

2.4. Cardiovascular measures

After the initial screening and anthropometric measures, participant sat on a comfortable chair for the assessment of cardiovascular variables at rest. First, heart rate (HR) data was continuously measured for 10 min with the participant seated in a comfortable position and silence. After this, systolic blood pressure (SBP) and diastolic blood pressure (DBP) was measured three times with 1-min interval using a digital blood pressure monitor (Omron, HEM-742int, Kyoto, Japan) on the left arm. Then, the heartbeat tracking task was performed (see description below).

Regarding blood pressure, the mean value of the three measures was used for the analysis. In addition, mean blood pressure (MAP) was calculated as: $MAP = [SBP + (2 \times DBP)] / 2$. All procedures for recording and analyzing HR and heart rate variability (HRV) data were performed following the methodology recommended by the Task Force [28]. First, a 5-min epoch with better data quality was extracted from the 10-min recording. This was performed by visual inspection of the data. The resting HR ($HR_{resting}$) was considered the mean value of the 5-min epoch. HRV represents a measure of the oscillation in the interval between consecutive heartbeats as well as the oscillations between consecutive instantaneous heart rates and represents a measure of cardiac autonomic control [28]. There are multiple HRV indexes that may be used. We analyzed the inter beats intervals (RR) and indexes of time domain such as the standard deviation of the normal-to-normal interval (SDNN), which reflects all the cyclic components responsible for variability, and square root of the mean squared differences of successive normal-to-normal intervals (RMSSD), which measures high-frequency variations in heart rate [28]. The greater SDNN and RMSSD the higher parasympathetic modulation on the heart [28]. On the frequency domain, we applied a Fast single-window Fourier Transform (FFT) over the R-R intervals. This decomposes the continuous data into its spectral components: (a) low-frequency band (LF, 0.04–0.15 Hz), which has been associated with both sympathetic and parasympathetic activity; (b) high-frequency band (HF, 0.15–0.40 Hz), which reflects the only parasympathetic activity; (c) and LF-HF ratio (LF/HF) as an index of sympathetic-vagal balance [28].

All measures of HR both at rest and during exercise were recorded using a heart monitor (RS800CX, Polar®, Finland) with an electrode attached to the participant's chest. The data were transferred to Polar Pro trainer 5® software for preliminary analysis and then analyzed in frequency and time domain in Kubios HRV® software.

2.5. Heartbeat tracking task

The subjects were assessed in a supine position and asked to avoid

Table 1
Sample characteristics. Mean and confidence interval [Lower bound – Upper bound].

Variables	Poor heartbeat perceivers (n = 15)	Good heartbeat perceivers (n = 17)	P	Hedges' g
Heartbeat perception accuracy (%)	63.9 [52.2–75.7]	95.3 [93.2–97.4]	0.001	2.38
Age (years)	25.1 [22.7–27.4]	22.9 [21.3–24.6]	0.11	–
Weight (kg)	72.1 [65.7–78.6]	72.2 [68.3–76.2]	0.99	–
Height (m)	1.71 [1.67–1.76]	1.73 [1.71–1.77]	0.37	–
Body mass index (kg/m ²)	24.4 [22.9–25.9]	23.8 [22.9–24.8]	0.50	–

any tactile manipulation that facilitated perception of heartbeats, such as positioning the fingers tips on the neck or forearm/wrist. Before the task, the subjects were instructed to relax and focus attention on their heartbeats. We used the heartbeat perception task protocol developed by Schandry [6]. Participants were requested to count their heartbeats between the start and stop signal of a soft tone, by concentrating on their heart activity. They were not permitted to check their pulse or to attempt any other physical manipulations that could facilitate the detection of heartbeats. Following stop tone, participants were asked to report the number of heartbeats counted. The task was performed three times, each with three random counting durations of 30, 45, and 60 s (i.e. total of nine trials) interspersed by 30-s interval. Participants received instructions on how to perform the test and were given some trials to familiarize them with the task. The test only started when participants fully understood the task. Their heartbeats were continuously recorded in a heart monitor as described above, away from the participant sight. Participants were not informed of the different counting length, duration or how many beats actually occurred. Individuals with heartbeat perception accuracy > 85% [29] were defined as good heartbeat perceivers ($n = 17$) and $\leq 85\%$ [7] as poor heartbeat perceivers ($n = 15$). Heartbeat perception accuracy was measured as follows [6]: $HPA = 1 - \{(\Sigma \text{ recorded heartbeats} - \Sigma \text{ counted heartbeats}) / \Sigma \text{ recorded heartbeats}\} \times 100\}$

2.6. Maximal incremental exercise test

The maximal incremental exercise test was performed in cycle ergometer (CG-04, Inbramed®, Porto Alegre, Brazil), with an initial load of 25 W and increment of 15 W per minute. Participants had to maintain 60–70 rotations per minute (rpm) until voluntary exhaustion occurred. The test was finished when the subjects could not maintain the specified cadence (< 5 rpm) for > 5 s. To be considered maximum individuals had to present at least two of the following criteria: (a) RPE ≥ 18 ; (b) achieve $\geq 90\%$ of predicted maximum HR (220-age); (c) inability to maintain the cadence despite strong verbal encouragement. All subjects reach met the criteria. Peak power was established as the highest power sustained in a complete 1-min stage. During the test, RR intervals were continuously recorded to calculate HRV_{TH} .

During incremental exercise, the decrease in HRV is not linear. HRV drops dramatically in the first stages up to approximately 50–60% of the peak power output, after which it tends to stabilize in values of approximately 2 ms [30,31]. This point represents the withdraw of parasympathetic cardiac modulation and is defined as the heart rate variability threshold (HRV_{TH}) [26,31]. There are different ways to calculate the HRV_{TH} with small differences among them [30,31]. The HRV_{TH} was considered in the present study as the stage (i.e., load in W) during the incremental test during which the first standard deviation (SD1) of RR intervals was lower than 3 ms [26,31]. Studies have validated this method against blood lactate and respiratory measures and confirmed that the HRV_{TH} is a non-invasive way to assesses the lactate and ventilatory threshold [26,31,32]. The individual file with continuous RR data during the maximum incremental test was analyzed and the 1-min of each stage of the incremental test was used to calculate SD1 (i.e., on an individual basis). The load, HR, and RPE at the stage in which HRV_{TH} was reached were registered.

Maximum HR (HR_{max}) was considered the highest value attained during the test. HR reserve (i.e., number of heartbeats per minute that an individual can achieve during exercise beyond their resting heart rate) was calculated as: $HR_{reserve} = HR_{max} - HR_{resting}$. Maximal oxygen uptake (VO_{2max}) was estimated using peak power (PP), body weight and age, applied in an equation developed specially for young males [33] as follows: $VO_{2max} = 0.518 + (0.01016 \times PP) + (0.01482 \times \text{Weight}) - (0.0292 \times \text{Age})$.

2.7. Rating of perceived exertion

RPE was defined as the subjective intensity of effort, strain, discomfort or fatigue that was felt during exercise [34]. The Borg 6–20 RPE scale translated and validated into Brazilian Portuguese was used to estimate whole-body perceived exertion during exercise and has shown good reliability and validity on healthy individuals against physiological measures, such as HR, blood lactate, oxygen uptake, ventilation, and respiratory rate [21]. The procedures for application of the Borg RPE scale was performed following the instructions by Robertson and Noble [35]. Before the maximum incremental test participants underwent a memory familiarization in which the RPE scale and what it measures was carefully explained, and examples related to the lowest and highest values were both given and required from the participant to confirm if they understood. RPE anchoring was: number 7 represents unloaded cycling while number 19 indicates an exertion similar to exhaustive cycling [36]. The RPE scale was displayed in front of the participants during the tests and instructions about reporting their perceived exertion were given before each test in both conditions. Participants were asked to accurately report the RPE last 15 s of each 1-min stage.

2.8. Statistics

Shapiro–Wilk normality test was used to test whether the data showed a normal Gaussian distribution. Accordingly, the data are reported as means $\pm$ standard deviation or median (interquartile range). An independent *t*-test was used if the data presented a normal distribution or Mann-Whitney test if otherwise to compare outcome variables between poor and good heartbeat perceivers. Outcome variables at rest were: $HR_{resting}$, RR interval, SDNN, RMSSD, LF, HF, LF/HF, systolic, diastolic and mean blood pressure. Outcome variables during exercise were: HR, power and RPE at HRV_{TH} as well as HR_{max} , $HR_{reserve}$, VO_{2max} and peak power. We calculated the effect size (ES) using Hedges' *g* and interpreted as follows: 0.20 = small; 0.50 = moderate; 0.80 = large [37]. Additionally, to test which variables were associated with interoceptive accuracy, it was also used Pearson's and partial correlation using the entire sample. A $P \leq .05$ was considered as significant.

3. Results

Individuals in both groups were homogenous concerning age and anthropometric characteristics (Table 1).

Table 2 displays cardiovascular measures at rest for both groups. Poor heartbeat perceivers exhibited greater resting heart rate, diastolic and mean blood pressure and lower RR interval, with no difference regarding the other variables. At rest, heartbeat perception accuracy was significantly correlated with $HR_{resting}$ ($r = -0.44$, $P = .02$) and RR interval ($r = 0.42$; $P < .03$), but not with systolic ($r = -0.07$; $P = .73$), diastolic ($r = -0.29$ ($P = .14$) or mean blood pressure ($r = -0.23$; $P = .25$).

On Table 3 the data regarding the exercise are displayed. Interestingly, poor and good heartbeat perceivers presented similar responses at the HRV_{TH} and maximum intensity. In addition, during exercise heartbeat perception accuracy was not correlated with peak power ($r = 0.18$; $P = .92$), VO_{2max} ($r = 0.11$; $P = .57$), $HR_{reserve}$ ($r = -0.03$; $P = .84$), power output at HRV_{TH} ($r = 0.21$; $P = .29$), or heart rate at HRV_{TH} ($r = -0.08$; $P = .68$). On the other hand, $HR_{resting}$ and RR interval at rest were correlated with peak power ($r = -0.49$, $P = .004$ and $r = 0.52$, $P = .002$), VO_{2max} ($r = -0.58$, $P = .001$ and $r = 0.56$, $P = .001$) and power output at HRV_{TH} ($r = -0.50$, $P = .003$ and $r = 0.52$, $P = .002$), respectively, showing that both the lower $HR_{resting}$ and longer RR interval (exhibited by good heartbeat perceivers at rest) were associated with better physical performance. Partial correlation between heartbeat perception accuracy and $HR_{resting}$ ($r = -0.47$, $P = .02$) and RR interval at rest ($r = 0.45$, $P = .02$), controlling for

Table 2

Comparison of measures of heart rate variability and blood pressure between poor ($n = 15$) and good ($n = 17$) heartbeat perceivers. Mean and confidence interval [Lower bound – Upper bound].

Variables	Poor heartbeat perceivers	Good heartbeat perceivers	<i>P</i>	Hedges' <i>g</i>
Resting heart rate (bpm)	76.1 [71.3–80.8]	69.7 [65.4–74.0]	0.04	1.06
RR interval (ms)	837.8 [757.4–918.3]	969.9 [908.2–1031.7]	0.009	0.97
SDNN (ms)	65.6 [45.6–85.7]	77.1 [60.8–93.5]	0.34	–
RMSSD (ms)	51.7 [27.5–76.0]	59.7 [44.1–75.4]	0.55	–
Low frequency (un)	64.2 [54.6–73.9]	60.4 [49.0–71.8]	0.59	–
High frequency (un)	35.8 [26.1–45.4]	39.6 [28.2–51.0]	0.59	–
Low frequency/High frequency (un)	2.9 [1.1–4.6]	2.6 [1.4–3.9]	0.81	–
Systolic blood pressure (mmHg)	125.4 [118.4–132.3]	119.6 [114.3–125.0]	0.16	–
Diastolic blood pressure (mmHg)	76.3 [69.7–82.9]	68.5 [65.2–72.0]	0.03	0.81
Mean blood pressure (mmHg)	92.7 [86.3–99.1]	85.6 [82.1–89.1]	0.04	0.75

Note: RR = intervals between ventricular systole (heartbeat); SDNN = standard deviation of the normal-to-normal RR intervals; RMSSD = the square root of the mean squared differences of successive normal-to-normal RR intervals.

physical fitness (i.e. VO_{2max}) showed that the correlation between interoceptive accuracy and cardiovascular measures at rest are independent of the fitness level.

4. Discussion

The purpose of this study was to assess the influence of interoceptive accuracy on cardiovascular measures at rest as well as RPE, cardiovascular and performance to a maximal intensity exercise. Results did not support the research hypothesis given that individuals with high and low interoceptive accuracy did not differ in any perceptual or physiological measure in submaximal (i.e., HRV_{TH}) and maximal exercise intensities. Interestingly, individuals with high interoceptive accuracy presented lower resting heart rate, longer inter-beat intervals, and lower diastolic and mean blood pressure.

Pollatos et al. [8] have suggested that there is a close relationship between sympathetic cardiovascular responses to physical stress and interoceptive accuracy. In this regard, habitually increased sympathetic outflow was suggested to be one variable contributing to the establishment of high interoceptive awareness [8]. However, our results did not confirm this assumption. First, there was no difference between individuals with high and low interoceptive accuracy for any of the sympathetic- or parasympathetic-related HRV indexes as well as systolic blood pressure. Secondly, if one would argue based on heart rate, inter-beat intervals, diastolic and mean blood pressure, the comparison showed that individuals with high interoceptive accuracy had lower resting heart rate, longer inter-beat interval, and lower blood pressure. These findings actually suggest a higher parasympathetic activity in the group with high interoceptive accuracy. Additionally, a negative correlation was found between heartbeat perception accuracy and resting heart rate, as well as the inter-beat interval (positive). However, there was no correlation between heartbeat perception and systolic or diastolic blood pressure. Therefore, this result suggests that higher interoceptive accuracy does not appear to be related to an increased sympathetic outflow as suggested previously [8].

Table 3

Comparison of measures of perceived exertion, physiological and physical performance during the maximal incremental test between poor ($n = 15$) and good ($n = 17$) heartbeat perceivers. Mean and confidence interval [Lower bound – Upper bound].

	Poor heartbeat perceivers	Good heartbeat perceivers	<i>P</i>
Peak power (W)	196.0 [177.5–214.5]	200.3 [188.0–212.6]	0.67
VO_{2max} ($ml \cdot kg^{-1} \cdot min^{-1}$)	39.6 [36.2–43.1]	40.9 [39.3–42.4]	0.48
Maximal heart rate (bpm)	182.5 [175.8–189.1]	168.4 [154.9–182.0]	0.06
Heart rate reserve (bpm)	107.2 [99.0–115.5]	99.1 [83.2–115.0]	0.34
Power at HRV_{TH} (W)	96.0 [82.9–109.1]	103.7 [95.2–112.2]	0.29
Heart rate at HRV_{TH} (bpm)	124.1 [118.7–129.5]	124.2 [114.1–134.2]	0.99
Rating of perceived exertion at HRV_{TH}	12.1 [11.1–13.2]	12.4 [11.7–13.2]	0.63

Note: VO_{2max} – maximum oxygen uptake; HRV_{TH} – heart rate variability threshold.

Additionally, Pollatos et al. [8] showed that individuals who responded with higher cardiovascular reactivity (e.g., heart rate, systolic blood pressure, stroke volume, cardiac output) to an isometric handgrip exercise also exhibited a greater degree of interoceptive awareness. Therefore, based on these findings it would be expected that during dynamic exercise, individuals with high interoceptive accuracy would show parasympathetic withdrawal at an early stage compared to individuals with low interoceptive accuracy, reaching HRV_{TH} before them. However, cardiac and physical load were similar in both groups at HRV_{TH} . Furthermore, once subjects with high interoceptive accuracy display better perception of body signals at rest, it also could be expected that RPE at HRV_{TH} to be different, but individuals of both groups displayed similar RPE. Therefore, at the HRV_{TH} , where the transition between light and moderate intensity occurs, interoceptive accuracy does not influence either cardiac, physical nor perceptual responses.

Contrary to previous studies with light to moderate intensity, there was also no difference in maximal exercise performance. Herbert et al. [14] assessed subjects in a 15 min cycling exercise, in which individuals were allowed to self-regulate cycling cadence (closed-loop exercise). They found that, in a setting in which subjects could choose their physical load, those with high interoceptive accuracy expended less physical effort (shorter covered distance and a smaller rise in heart rate, stroke volume, and cardiac output) than individuals with low interoceptive accuracy, although with similar RPE [14]. Additionally, Pennebaker and Lightner [13] evidenced that when exercise intensity was constant, individuals listening to the amplification of their breathing reported higher RPE than when they listened to distracting sounds (street sounds: people walking, idle conversations, sequence of radio talk show and music, parts of a lecture and movie, cars driving by, etc., for up to 20 s each vignette) or no sound. Then, on a second experiment, individuals ran faster on the cross-country course with the same level of RPE compared to when running on a lap course [13]. Given that when individuals are allowed to self-regulate exercise intensity they seem to do so based on their RPE and when they cannot self-regulate, those with amplified body signals (i.e. breathing) exhibit higher RPE, it would be

more likely that subject with high and low interoceptive would perceive maximal incremental exercise differently and exhibit distinct maximal performance. A possible reason could be due to the fact that the afferent signals ascending to the brain and are re-represented in the AIC, where it is processed into a subjective evaluation of interoceptive state. Then this information is forward to the orbitofrontal cortex, where hedonic valence is represented that will influence emotion, mood motivation, and consciousness [1,2]. How, when and to what extent individuals with low and high interoceptive accuracy differ in the re-representation of interoceptive signals is unknown. However, maximal power output, relative and absolute cardiac load were comparable between individuals with high and low interoceptive accuracy, refuting this assumption. Additionally, there was no correlation between heartbeat perception and any physical or cardiac measure during maximal exercise. Therefore, interoceptive accuracy does not influence maximal exercise performance.

One possible explanation for this result is that acute increases in exercise intensity leads to improvements in interoceptive accuracy [38,39]. Schandry, Bestler, and Montoya [38] performed an experiment in which individuals were positioned in different body positions on a tilt table and performed exercise with different intensities (i.e., unloaded cycling, 25 W, 50 W, 75 W). They found that interoceptive accuracy increased linearly with increasing exercise intensity and more horizontal body positions [38]. The mechanism for increased interoceptive accuracy due to augmented exercise intensity was that greater heart muscular activity leads to increases in momentum (i.e., acceleration of stroke volume through the vessels), generating larger pressure and/or stretch stimuli in the heart tissue and/or the surrounding regions [38]. In fact, our results showed that individuals' heart rate was increased by > 100 beats per minute by the end of the test, confirming the increased cardiovascular load. Additionally, Edwards and Polman [16] argue that the level of awareness through the processing of afferent feedbacks depend on the strengthening of these signals as well as the exercise intensity. Furthermore, Ekkekakis [40–42] proposed the dual model theory, which postulates that during exercise at an intensity below the ventilatory threshold, cognitive processes, such as self-efficacy, self-presentational concerns, goals, and attributions, predominantly influence psychophysiological responses. By contrast, at an intensity equal or above the ventilatory threshold, the psychophysiological responses are influenced primarily by interoceptive signals, such as muscular, temperature, respiratory, and acidosis [40–42]. The homogeneous psychophysiological responses at high intensities, therefore, would be the result of increased afferent feedback signaling [40–42]. Therefore, the intensified interoceptive cues, as well as enhanced interoceptive accuracy in both groups, may have led to similar processing of afferent peripheral feedbacks to all individuals and, consequently, the same exercise performance, independently of their interoceptive accuracy at rest.

As it has been recently proposed, numerous factors may influence the bidirectional influence between top-down and bottom-up systems [3,43]. In this regard, the attentional focus plays an important role in exercise regulation [44]. Interestingly, studies have tried to manipulate the attentional focus to divert the exerciser attention from the interoceptive signals in order to improve psychophysical responses to exercise and/or exercise performance [45–48]. The attentional focus may be defined as associative (i.e., focus on bodily sensations, such as respiration, temperature, and muscular fatigue) or dissociative (i.e., cognitive 'blocking out' sensations of pain or discomfort related to physical effort) [45]. Strategies to manipulate the attentional focus involve dissociative stimulus, use of audiovisual stimuli (e.g., music, TV), self-talk, boosting self-efficacy, among others [45]. Possible explanations involves the competition of cues that proposes the exercise experience will depend on whether an individual maintains either an internal or external attentional focus [45]. Another possibility is the parallel processing of information model that proposes that behavioral responses depend on what stimulus are attended to and come into focal

awareness, in other words, if you focus on your breathing altered by exercise you may feel more tired than if focus elsewhere [45]. The social psychophysiological model proposes that in low to moderate exercise intensities psychological factors contribute to a greater extent to perceived exertion (e.g., cognitive strategies, individual differences, motivation), thus focusing on their breathing decreases the influence of cognitive factors and increases RPE [45]. So far, however, the results of studies using dissociative strategies are contradictory. A recent study reported that even though a dissociative strategy (i.e., self-talking) did not improved ultramarathon performance, ultramarathon runners found this dissociative strategy helpful during competition and most of them continued to use that strategy 6 months after the study, particularly during competition [46].

Some limitations should be considered when generalizing the results of this study. The fact that this study involved only men leaves an open question regarding whether women show the same responses as men do. However, the assessment of only men was done in order to eliminate the variance in interoceptive accuracy due to gender differences [49]. The absence of measures of cardiac contractility, stroke volume and momentum [38] does not allow the establishment of mechanisms, although it was not the purpose of this study. Future studies should take these into account and also include measures of interoceptive accuracy during exercise in order to test if and how different exercise intensities change interoceptive accuracy and how it relates with individual perception (e.g., how much the body signal bothers you during exercise or how do you cope with it). In addition, it still unknown if and how interoceptive accuracy influences individuals perceptions (e.g. RPE, affective responses, attentional focus, etc.) and physiological responses during a recommended bout of exercise (e.g., 30 min of moderate intensity or 20 min of vigorous intensity).

5. Conclusion

In summary, at rest individuals with higher interoceptive accuracy presented lower HR, longer RR intervals, lower diastolic and mean blood pressure as compared to the lower interoceptive accuracy group, but no other measure of cardiac autonomic control was different between groups. During dynamic exercise, interoceptive accuracy does not influence performance, physiological, perceptual responses at the submaximal or maximal intensities in an incremental exercise test. This result suggests that the influence of interoceptive accuracy on the physical, physiological and perceptual responses to exercise might be dependent upon exercise intensity, mode (e.g., dynamic, static) and design (e.g., constant or incremental load; self-paced or imposed intensity). Also, only one dimension of interoception was assessed (i.e., interoceptive accuracy) with a single measure (i.e., heartbeat tracking task) in cycling exercise. Future research should consider other body signal and/or use multiple forms to measure the different dimensions of interoception (i.e., accuracy, sensibility, and awareness).

Acknowledgments

The authors would like to thank the following institutions for providing scholarship to the authors: Rio Grande do Norte Research Foundation (FAPERN) - master's degree student to DGSM; National Council for Scientific and Technological Development (CNPq) - scientific initiation to MPMT and master's degree student to LFFJ; Coordination for the Improvement of Higher Education Personnel (CAPES) - scientific initiation to PHDN and master's degree student to PMDA and AIF.

Conflict of interest

The authors declare no conflict of interest

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